JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR
COLLEGE OF ENGINEERING (AUTONOMOUS)ANANTHAPURAMU– 515 002 (A.P) INDIA

B. Tech. – I Year I Semester
(Common to EEE, ECE, CSE)

Subject code	Title of the Subject	L	T	P	C
23A15301	**Chemistry**	**3**	**0**	**0**	**3**

Course Objectives:

- To familiarize engineering chemistry and its applications
- To train the students on the principles and applications of electrochemistry and polymers
- To introduce instrumental methods, molecular machines and switches.

Course Outcomes:

- **CO1:** Apply Schrodinger wave equation to hydrogen atom, Illustrate the molecular orbital energy level diagram of different molecular species, Explain the band theory of solids for conductors, semiconductors and insulators Discuss the magnetic behavior and colour of complexes.
- **CO2:** Explain the band theory of solids for conductors, semiconductors and insulators. Explain Basic concept and application of Super Conductors Super Capacitors Demonstrate the application of Fullerenes, carbon nano tubes and Graphines nanoparticles
- **CO3:** Apply Nernst equation for calculating electrode and cell potentials, differentiate between potentiometric and conductometric titrations, Explain the theory of construction of battery and fuel cells, solve problems based on cell potential
- **CO4:** Explain the different types of polymers and their applications, Explain the preparation, properties and applications of PVC, Bakelite Describe the mechanism of conduction in conducting polymers, Discuss Buna-S and Buna-N elastomers and their applications
- **CO5:** Explain the different types of spectral series in electromagnetic spectrum, Understand the principles of different

MAPPING BETWEEN COURSE OUTCOMES AND PROGRAMME OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

UNIT I: Structure and Bonding Models: (10 hrs.)

Fundamentals of Quantum mechanics, Schrodinger Wave equation, significance of Ψ and Ψ^2, particle in one dimensional box, molecular orbital theory – bonding in homo- and hetero nuclear diatomic molecules – energy level diagrams of O_2 and CO, etc. π-molecular orbitals of butadiene and benzene, calculation of bond order.

Learning Outcomes:
At the end of this unit, the students will be able to

- **apply** Schrodinger wave equation to hydrogen atom (L3)
- **illustrate** the molecular orbital energy level diagram of different molecular species (L2)
- **explain** the calculation of bond order of O_2 and CO molecules (L2)
- **Discuss** the basic concept of molecular orbital theory (L3)

UNIT II: Modern Engineering materials (8hrs)

Semiconductors – Introduction, basic concept, application
Super conductors-Introduction, basic concept, applications.
Super capacitors: Introduction, Basic Concept-Classification – Applications.
Nano materials: Introduction, classification, properties and applications of Fullerenes, carbon nano tubes and Graphines nanoparticles.

Learning Outcomes:
At the end of this unit, the students will be able to

- **Explain** the Applications of Super Capacitors (L2).
- **Discuss** the Basic concept of Super Conductors (L3).
- **Explain** the band theory of solids for conductors, semiconductors and insulators (L2)
- **Demonstrate** the application of Fullerenes, carbon nanotubes and Graphines nanoparticles (L2).

UNIT III: Electrochemistry and Applications (10hrs)

Electrochemical cell, Nernst equation, cell potential calculations and numerical problems, potentiometry-potentiometric titrations (redox titrations), concept of conductivity, conductivity cell, conductometric titrations (acid-base titrations), pH metry.
Electrochemical sensors – potentiometric sensors with examples, amperometry sensors with examples.
Primary cells – Zinc-air battery, Sodium-Air battery Secondary cells –lithium-ion batteries- working of the batteries including cell reactions;
Fuel cells, hydrogen-oxygen fuel cell– working of the cells. Polymer Electrolyte Membrane Fuel cells (PEMFC).

Learning Outcomes:
At the end of this unit, the students will be able to

- **apply** Nernst equation for calculating electrode and cell potentials (L3)
- **differentiate** between potentiometric and conduct metric titrations (L2)
- **explain** the theory of construction of battery and fuel cells (L2)
- **solve** problems based on cell potential (L3)

UNIT IV: Polymer Chemistry (10hrs)

Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation, Polydispersity index (PDI)-significance
Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of – PVC, Teflon, Bakelite, Nylon-6, 6, carbon fibers.
Elastomers–Buna-S, Buna-N–preparation, properties and applications.
Conducting polymers – polyacetylene, polyaniline, – mechanism of conduction and applications. Bio-Degradable polymers - Poly Glycolic Acid (PGA), Polyl Lactic Acid (PLA).

Learning Outcomes:

At the end of this unit, the students will be able to

- **explain** the different types of polymers and their applications (L2)
- **explain** the preparation, properties and applications of Bakelite, Nylon-6,6, and carbon fibers (L2)
- **describe** the mechanism of conduction in conducting polymers (L2)
- **discuss** Buna-S and Buna-N elastomers and their applications (L2)

UNIT V: Instrumental Methods and Applications (10 hrs.)

Electromagnetic spectrum. Absorption of radiation: Beer-Lambert's law. UV-Visible Spectroscopy, electronic transition, Principle, Instrumentation and Applications, IR spectroscopy, fundamental modes and selection rules, Principle, Instrumentation and Applications. Chromatography-Basic Principle, Classification-HPLC: Principle, Instrumentation and Applications.

Learning outcomes:

After completion of this unit, students will be able to:

- **explain** the different types of spectral series in electromagnetic spectrum (L2)
- **understand** the principles of different analytical instruments (L2)
- **explain** the different applications of analytical instruments (L2)

Textbooks:

1. Jain and Jain, Engineering Chemistry, 16/e, Dhanpat Rai, 2013.
2. Peter Atkins, Julio de Paula and James Keeler, Atkins' Physical Chemistry, 10/e, Oxford University Press, 2010.

Reference Books:

1. Skoog and West, Principles of Instrumental Analysis, 6/e, Thomson, 2007.
2. J.D. Lee, Concise Inorganic Chemistry, 5th Edition, Wiley Publications, Feb.2008
3. Textbook of Polymer Science, Fred W. Bill mayer Jr, 3rd Edition